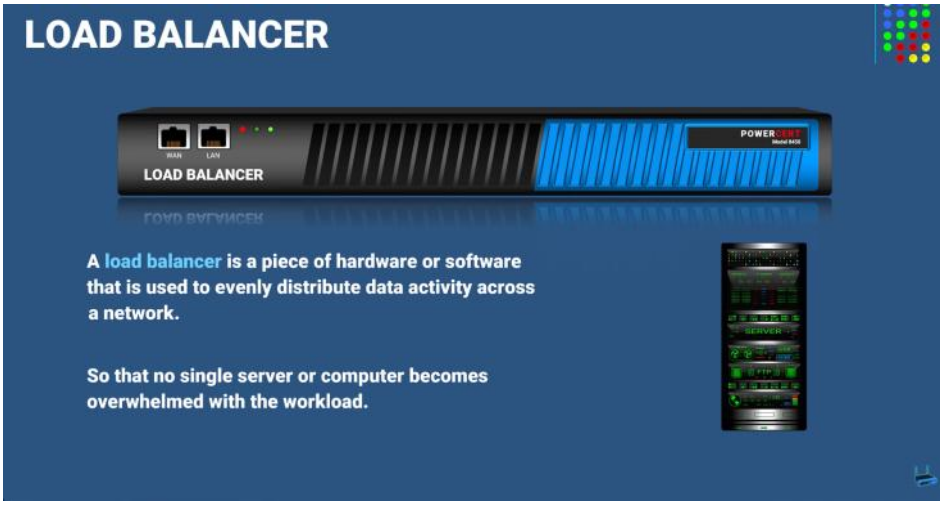
Load Balancer

Tuesday, March 31, 2026 10:49 AM

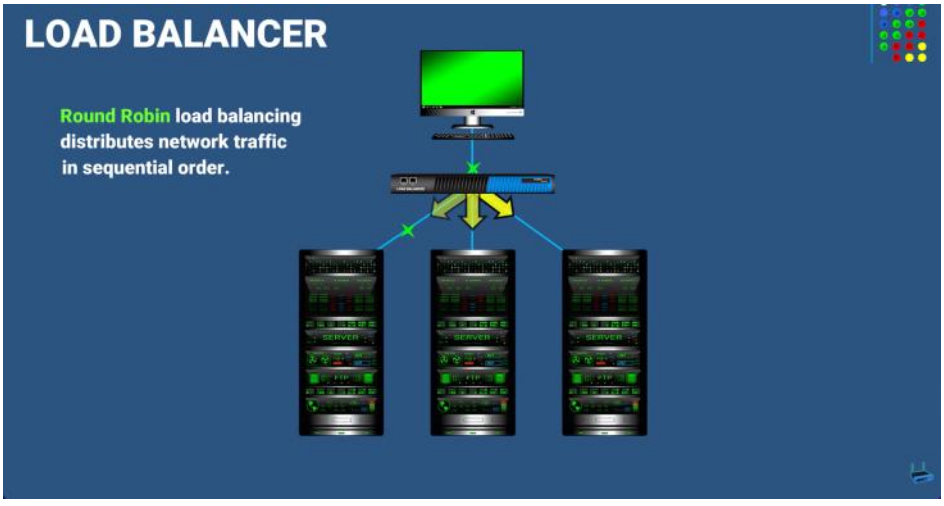
A load balancer is hardware or software used to distribute incoming network traffic across multiple servers. This ensures that no single server becomes overwhelmed, improving the performance, reliability, and availability of applications.



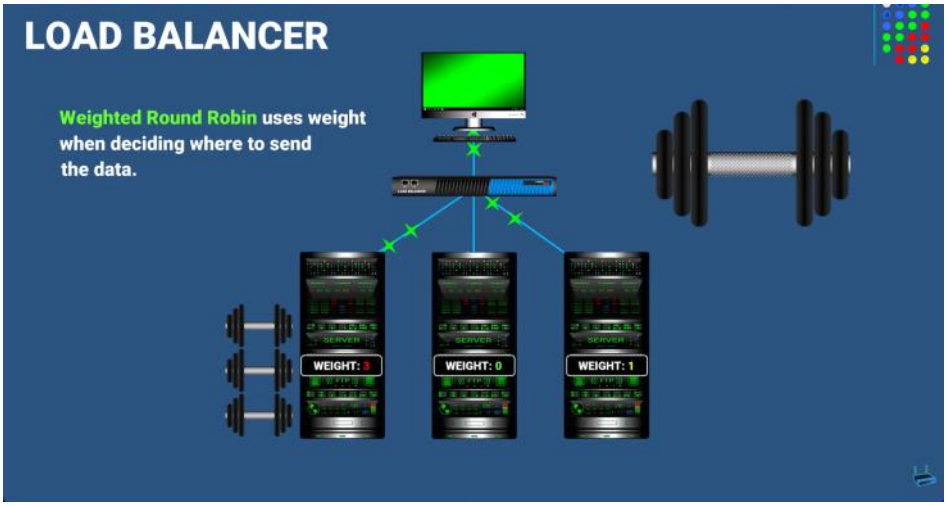
Load Balancing Algorithms

Load balancers use specific rules to determine where to send traffic:

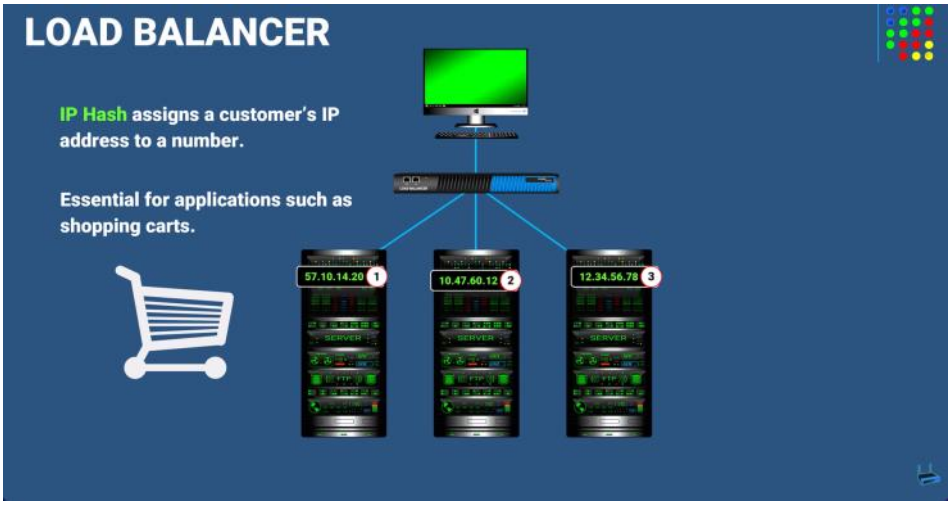
1. Static Algorithms: Follow fixed rules regardless of server status.



Round Robin: Distributes requests sequentially to servers. It will give the orders to the servers in sequential manner, making it easy for the servers to manage request overload equally rather than giving all the workload to a single server. It can be used when all the servers have same capacity.

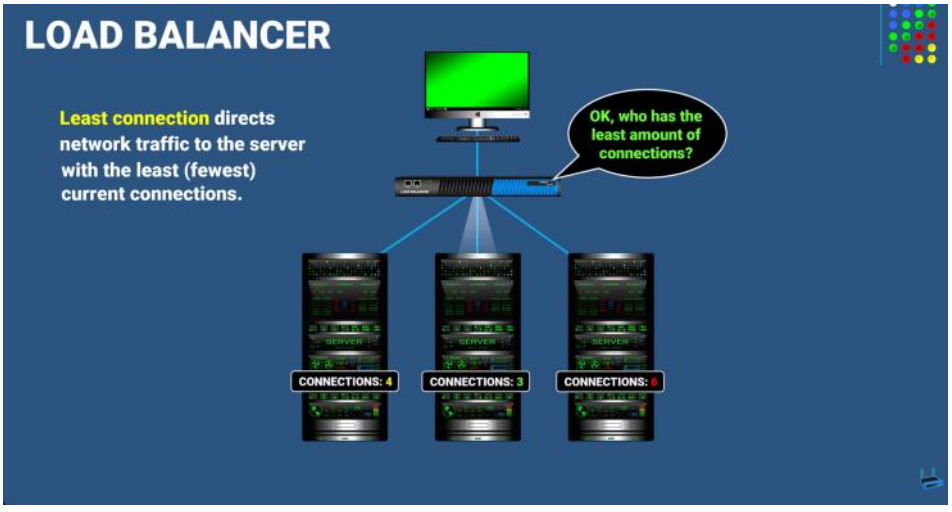


Weighted Round Robin: Assigns traffic based on server capacity, sending more to stronger servers.

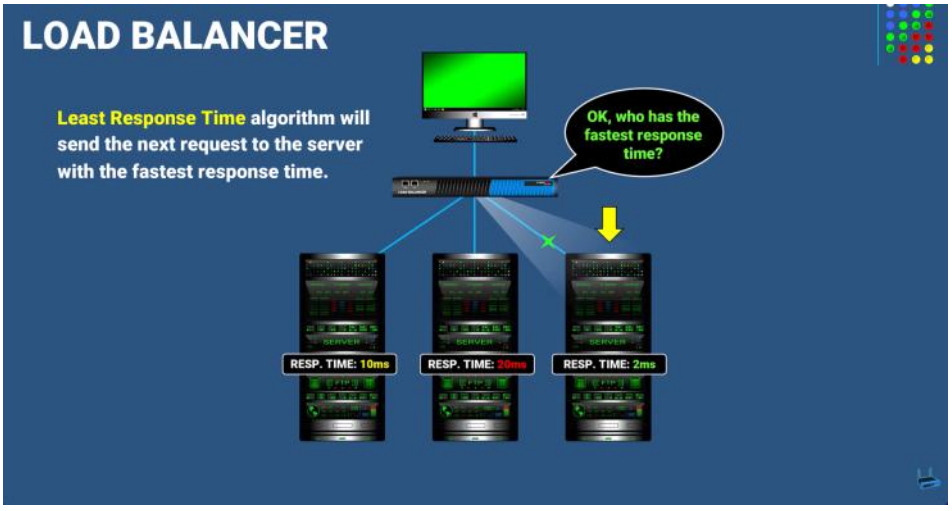


IP Hash: Routes traffic based on the client's IP address, ensuring users stay connected to the same server for consistent sessions (like shopping carts).

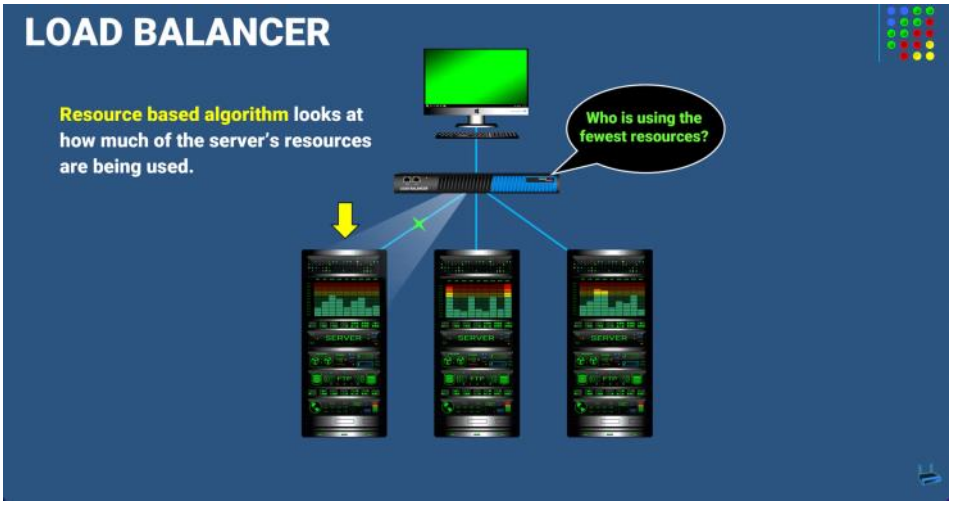
2. Dynamic Algorithms: Monitor server health and workload in real-time.



Least Connection: Sends new requests to the server with the fewest active connections.



Least Response Time: Directs traffic to the server that processes requests the fastest.



Resource-based: Uses agents to track CPU and RAM usage, choosing the server with the most available resources.

Key Benefits

- Performance: Reduces lag and response times by balancing the workload.
- High Availability: Prevents outages by rerouting traffic if a server fails.
- Scalability: Allows for easy addition or removal of servers as demand changes.
- Security: Helps defend against DDoS attacks by monitoring traffic or distributing the attack volume across multiple servers.